

Jubair Ahmed Nabin

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Summary

Lecturer and researcher in Computer Science & Engineering at IUBAT, Dhaka. Research focuses on federated learning, intrusion detection systems, explainable AI (SHAP/LIME), adversarial machine learning, and video anomaly detection. Author of 7 peer-reviewed papers at IEEE and Springer venues.

Education

B.Sc. Chittagong University of Engineering & Technology (CUET), Bangladesh, Computer Science and Engineering Sept 2018 – June 2023

- CGPA: 3.64 / 4.00
- Relevant Coursework: Machine Learning, Data Structures & Algorithms, Computer Networks, Database Management Systems, Artificial Intelligence, Software Engineering

Experience

International University of Business Agriculture and Technology (IUBAT), Lecturer, Department of CSE Dhaka, Bangladesh
Jan 2023 – present
3 years 10 months

- Courses: Artificial Intelligence, Data Science, Data Structures, Database Management Systems, System Analysis & Design
- Mentored 35+ undergraduate students on ML and cybersecurity research projects; two student-led projects advanced to peer-reviewed publication (IEEE PECCII 2026, IEEE ICMI 2026)

CUET / IUBAT, Researcher — Machine Learning & Cybersecurity Dhaka, Bangladesh
Jan 2022 – present
4 years 10 months

- Developed a privacy-preserving CNN-LSTM architecture within a Federated Learning framework for network intrusion detection, achieving over 99.36% accuracy across all seven attack classes on CIC-IDS2017 without centralizing raw data; raised Web Attack F1-score from 0.20 (CNN baseline) to 0.93, outperforming LSTM, CNN+BiLSTM, and CNN+GRU variants
- Built a novel file-entropy dataset and detection pipeline for crypto-ransomware, developing a custom FUSE-based filesystem to intercept file write/delete operations across 123 file types (30 ransomware strains, 93 benign); trained ML models achieving up to 96% detection accuracy (first-author, Springer LNNS 2025)
- Evaluated cross-domain adversarial robustness and XAI explanation stability in phishing detection, comparing PGD and Carlini-Wagner attack accuracy alongside SHAP explanation drift across feature-engineered vs. raw-URL input representations (IEEE PECCII 2026)
- Designed a two-stage CNN + Vision Transformer pipeline for video anomaly detection on the UCF-Crime dataset — 98% accuracy on binary classification, 95% on multi-class categorization; introduced a keyframe extraction algorithm reducing per-video processing time to 20ms

Publications

Cross-Domain Adversarial Robustness and Explainability Stability in Phishing Detection Systems June 2026

Md. Rifat Hassan, **J. A. Nabin**, Toyeer-E-Ferdoush
[10.1109/PECCII70991.2026.11661863](https://doi.org/10.1109/PECCII70991.2026.11661863) (IEEE International Conference on Power, Electronics, Communications, Computing, and Intelligent Infrastructure (PECCII))

Unraveling the Factors of Procrastination: An Insightful Assessment of Procrastination Levels Through ML and XAI Mariam Sarker, Anwar Hossain Efat, Maria Afrin Khan, Minhaz Zibran, J. A. Nabin 10.1109/ICMI68585.2026.11539948 (IEEE 5th International Conference on Computing and Machine Intelligence (ICMI))	Apr 2026
A Novel File Entropy Dataset for Crypto-Ransomware Detection Using Machine Learning J. A. Nabin , Md Mokammel Haque 10.1007/978-981-96-2721-9_20 (2nd International Conference on Machine Intelligence and Emerging Technologies (Springer LNNS))	July 2025
Federated Deep Learning for Cybersecurity and Intrusion Detection in Decentralized Networks Naeem Mia, J. A. Nabin , Suzad Mohammad, Mahedi Hasan, Fahim Shakil Tamim, Dipta Mohon Das 10.1109/ICAECA63854.2025.11012607 (IEEE 3rd International Conference on Advancements in Electrical, Electronics, Communication, Computing and Automation (ICAECA))	July 2025
A Novel Temporal-Aware Adaptive Feature Selection Strategy for Network Intrusion Detection Systems Naeem Mia, Md Mahfuzul Haque Gazi, J. A. Nabin , Fahim Shakil Tamim, Suzad Mohammad, Md Rownak Ul Islam 10.1109/GINOTECH63460.2025.11076711 (IEEE Global Conference in Emerging Technology (GINOTECH))	May 2025
Two-Stage Vision Transformer-Based Framework for Anomaly Detection and Classification in Surveillance Videos Mahedi Hasan, J. A. Nabin , Naeem Mia, Fahim Shakil Tamim, Suzad Mohammad, Dipta Mohon Das 10.1109/ECCE64574.2025.11013374 (IEEE International Conference on Electrical, Computer and Communication Engineering (ECCE))	Feb 2025

Technical Skills

Programming Languages: Python, C++, SQL, JavaScript
ML / DL Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn
Federated Learning: FL aggregation pipelines, privacy-preserving distributed training
Explainability & Security: SHAP, LIME, adversarial robustness (PGD, Carlini-Wagner)
Data & Visualization: Pandas, NumPy, Matplotlib, OpenCV
Tools & Environments: Git, Linux, LaTeX, Jupyter Notebook, VS Code, Google Colab

Awards & Achievements

2026: Miyan Publications Reward Ceremony — Research Publication Recognition
2018 – 2022: CUET Talent Hunt IUPC — 5th Position
2020, 2022: Meta Hacker Cup — Round 1 Qualifier
2015, 2017: General Board Scholarship

Conference Presentations

- IEEE PECCII 2026
- IEEE ICMI 2026
- IEEE ICAECA 2025
- IEEE GINOTECH 2025
- IEEE ECCE 2025
- Springer MIET 2025